

16-831: Introduction to Robot Learning, Fall 2024

Preference-based Learning (Part II) and Exploration



Instructor: Guanya Shi

Recording Reminder

- ☐ This session will be audio recorded for educational use by other students in this course.

Guest Lectures

❑ Please attend in person. “Almost” mandatory (strongly encourage)!



Lecture 18, 10/31 (Thu)
Aviral Kumar, CMU CSD/MLD

Offline RL: Algorithms and Applications



Lecture 22, 11/19 (Tue)
Yonatan Bisk, CMU LTI

Foundation Models in Robotics

Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☐ Topic group 2: Machine/Deep learning refresher
- ☐ Topic group 3: Imitation learning
- ☐ Topic group 4: Model-free reinforcement learning
- ☐ Topic group 5: Model-based reinforcement learning
- ☒ Topic group 6: Bandit, preference-based learning, exploration
- ☐ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☐ Topic group 8: Specialized topics (one guest lecture)
- ☐ Project presentation

Outline

- ❑ Recap: Bandits
- ❑ Continue: Preference-based learning

- ❑ Exploration in RL: Motivations
- ❑ Exploration via intrinsic reward:
 - Curiosity
 - Skill discovery
 - Random network distillation

(Some slides are from 22F, 23F, Berkeley CS 285, CMU 10-703, and Amy Kejun Li)

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Multi-Arm Bandits (MAB) Problem: Formulations

- ❑ k actions (a.k.a. arms or bandits)
- ❑ Each action has an average reward: $\mu_1, \mu_2, \dots, \mu_k$
 - Unknown to us
 - μ^* is the best
- ❑ For $t = 1, \dots, T$
 - Algorithm chooses action $a(t) \in [1, k]$
 - Receives random reward $R(t)$
 - Expectation of $R(t)$ is $\mu_{a(t)}$

Algorithm only receives
feedback on chosen action

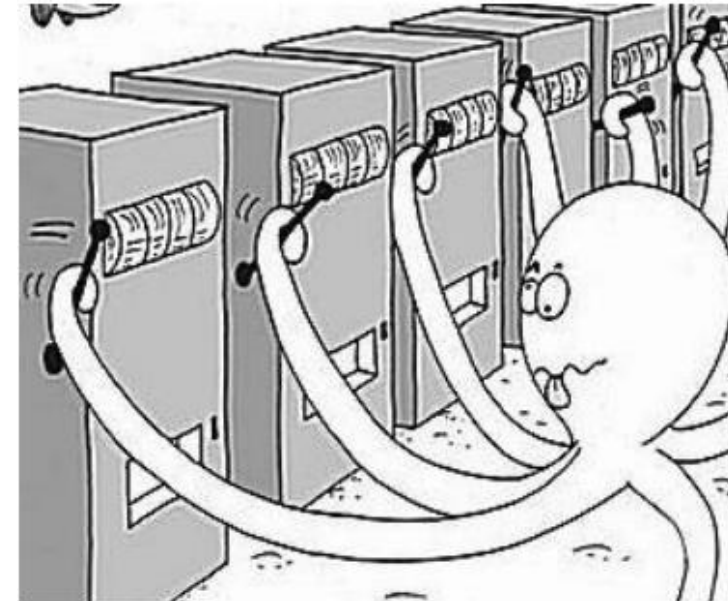
- ❑ **Goal:** minimize **Regret** = $T\mu^* - (\mu_{a(1)} + \mu_{a(2)} + \dots + \mu_{a(T)})$

If we had perfect information to start

Expected reward of algorithm

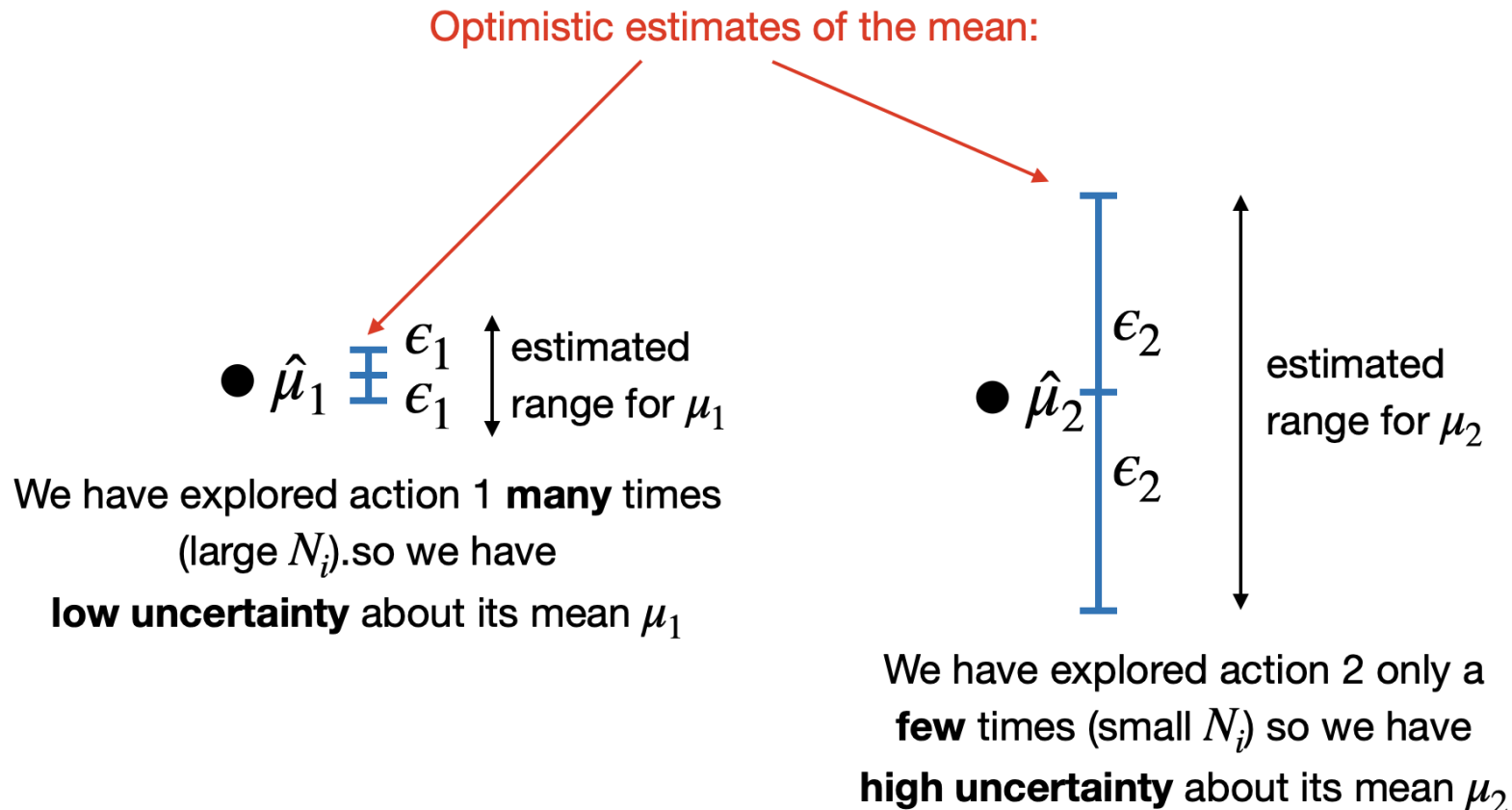
- ❑ Why this problem is hard?

- Counterfactual reasoning (we only have feedback from chosen action)
- Randomness (we have to balance between exploration and exploitation)



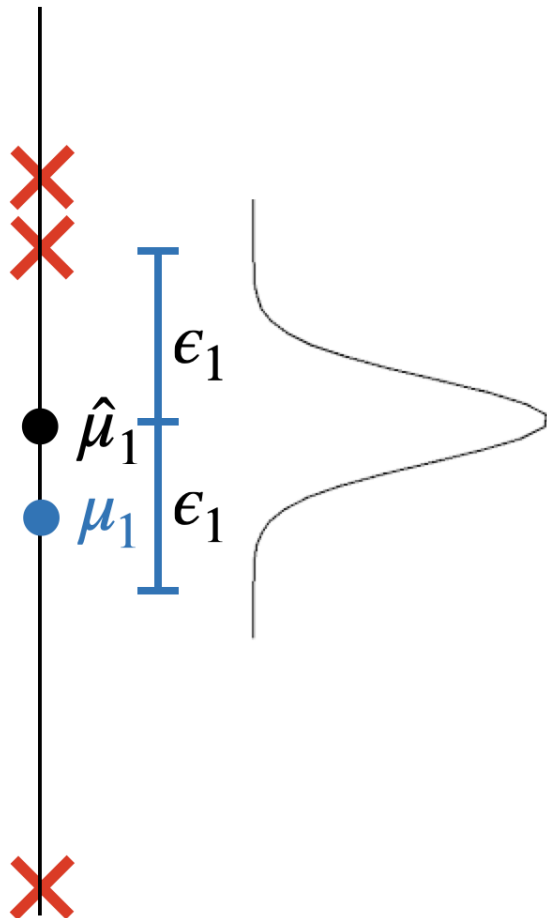
UCB: Upper Confidence Bounds

- ❑ Key idea: choose the most optimistic action, based on $\hat{\mu} + \epsilon$
- ❑ Principles: **optimism in the face of uncertainty** (OFU)
- ❑ UCB will choose action 2 (optimize the optimistic estimates)!



Thompson Sampling

- ❑ UCB: estimate a range that the mean lies within, with high probability
- ❑ Thompson sampling: estimate a posterior distribution over μ (NOT a range)



Thompson Sampling does the following:

1) Estimate the posteriors for each arm:

$$p(\mu_i | r_1, \dots, r_n) = \mathcal{N}(\mu'_i, \sigma'_i) \text{ where } r_1, \dots, r_n \text{ are only the rewards for arm } i$$

2) For each arm, sample a mean from its posterior distribution:

$$\hat{\mu}_i \sim \mathcal{N}(\mu'_i, \sigma'_i)$$

3) Choose the arm with the highest **sampled** mean

4) Take the action, observe the reward and go back to step 1

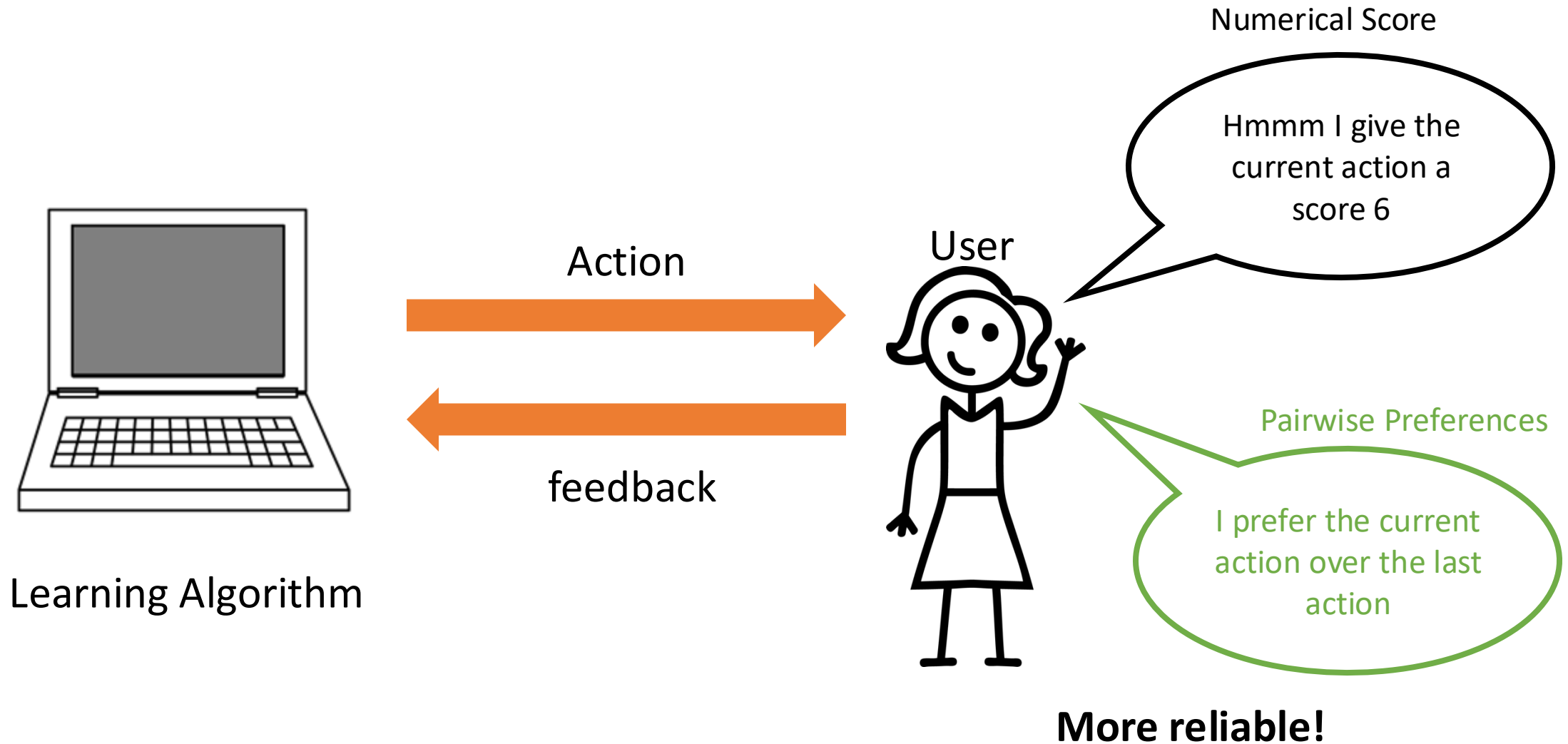
It is why it's also
unknown as
posterior sampling

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What is Preference-based Feedback?



Why Preference-based Learning

❑ Preference-based feedback is easier to obtain and often more reliable!

Preference-Based Learning for Exoskeleton Gait Optimization

Maegan Tucker^{*1}, Ellen Novoseller^{*2}, Claudia Kann¹,
Yanan Sui³, Yisong Yue², Joel W. Burdick^{1,2}, and Aaron D. Ames^{1,2}

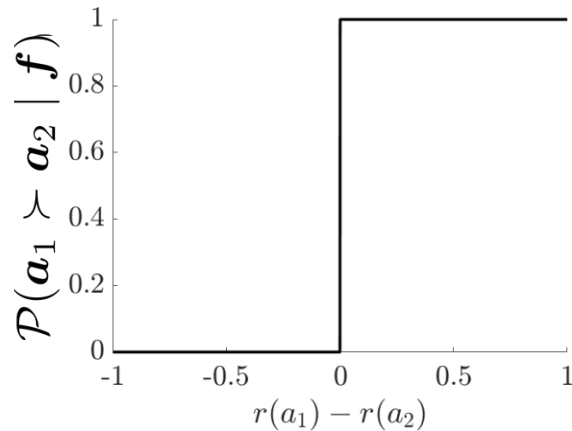


Modeling User Preferences

- Assumption: each action a has a corresponding underlying utility $f(a)$
 - f is unknown and to be learned

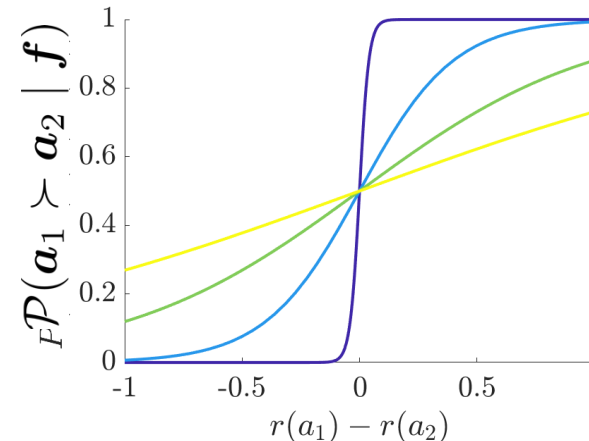
Ideal Noise-Free Preferences

$$\mathcal{P}(a_1 \succ a_2 \mid f(a_1), f(a_2)) = \begin{cases} 1 & \text{if } f(a_1) \geq f(a_2), \\ 0 & \text{otherwise,} \end{cases}$$

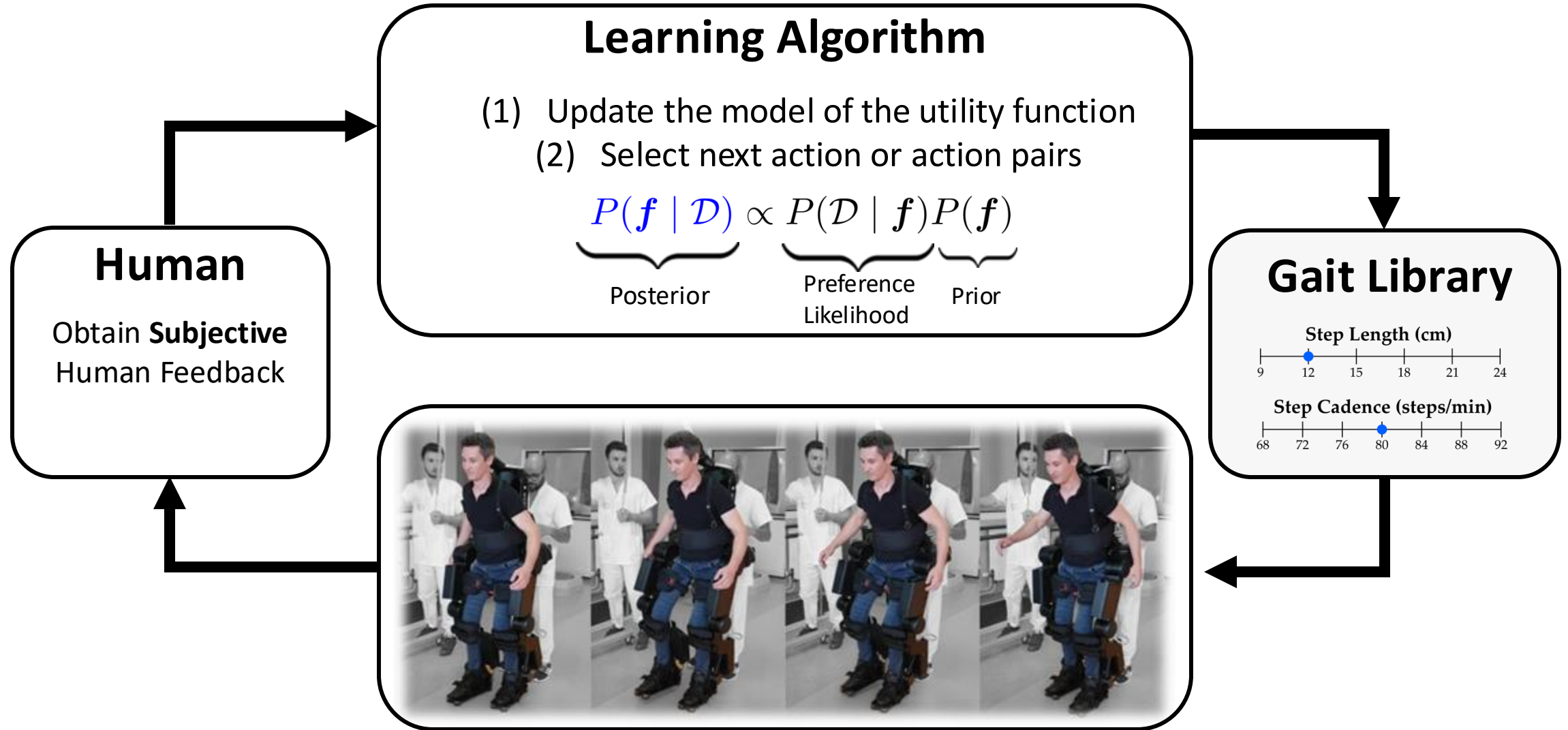


Noisy Preferences

$$\mathcal{P}(a_1 \succ a_2 \mid f(a_1), f(a_2)) = g\left(\frac{f(a_1) - f(a_2)}{c_p}\right)$$



The Learning Pipeline

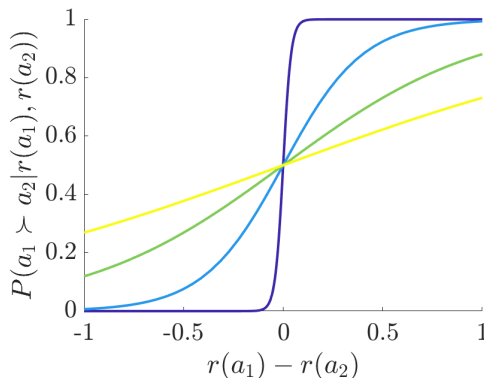


How to Design and Learn the Utility Function $f(a)$

$$\underbrace{P(\mathbf{f} \mid \mathcal{D})}_{\text{Posterior}} \propto \underbrace{P(\mathcal{D} \mid \mathbf{f})}_{\text{Preference Likelihood}} \underbrace{P(\mathbf{f})}_{\text{Prior}}$$

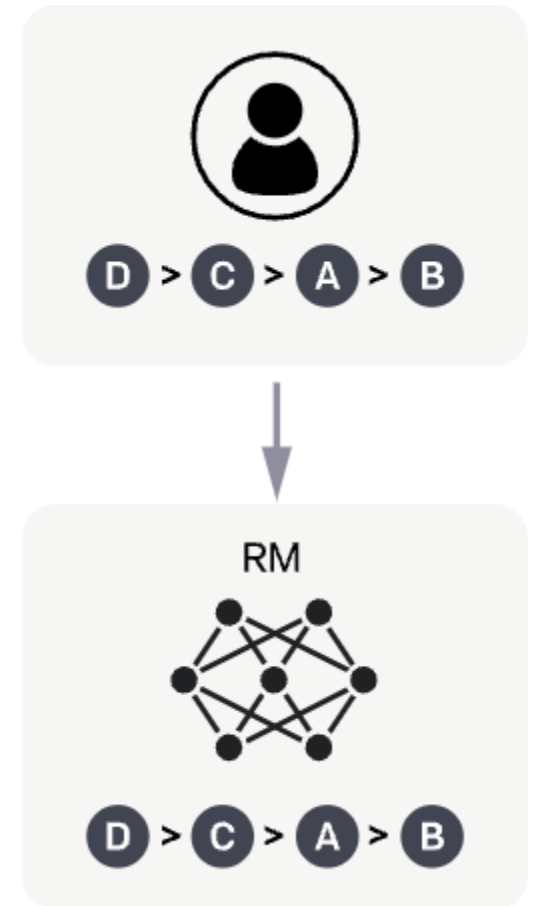
- ❑ $\mathcal{D}$: collected data ($a_1 \succ a_2, \dots$)
- ❑ f : utility model
 - Learned by maximizing likelihood
 - Might depend on context s : $f(s, a)$
 - Example: Gaussian process, LLM

$$P(a_1 \succ a_2 \mid f(a_1), f(a_2)) = g\left(\frac{f(a_1) - f(a_2)}{c_p}\right)$$




Explain reinforcement learning to a 6 year old.

- | | |
|--|--|
| A
In reinforcement learning, the agent is... | B
Explain rewards... |
| C
In machine learning... | D
We give treats and punishments to teach... |



How to Select the Next Action

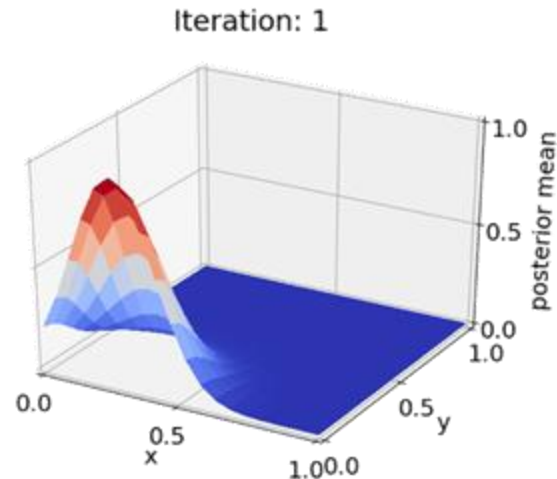
- ❑ Two different formulations: active learning and Bayesian optimization

Active Learning

Goal: Discover true utilities f

Select actions to: Maximize accuracy
(Max Entropy, Information Gain)

E.g., learn black box model

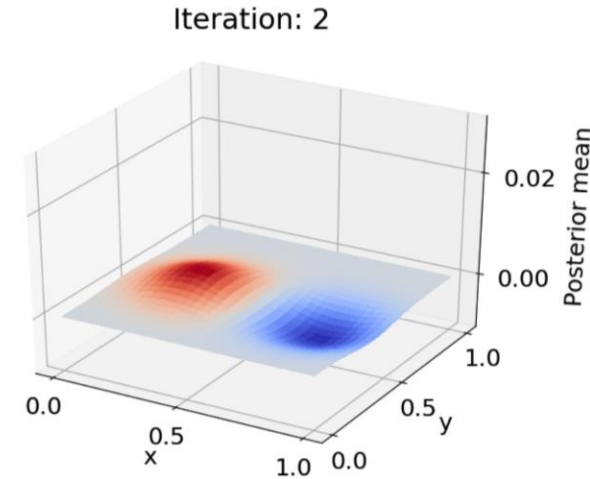


(Bayesian) Optimization

Goal: Select the best action

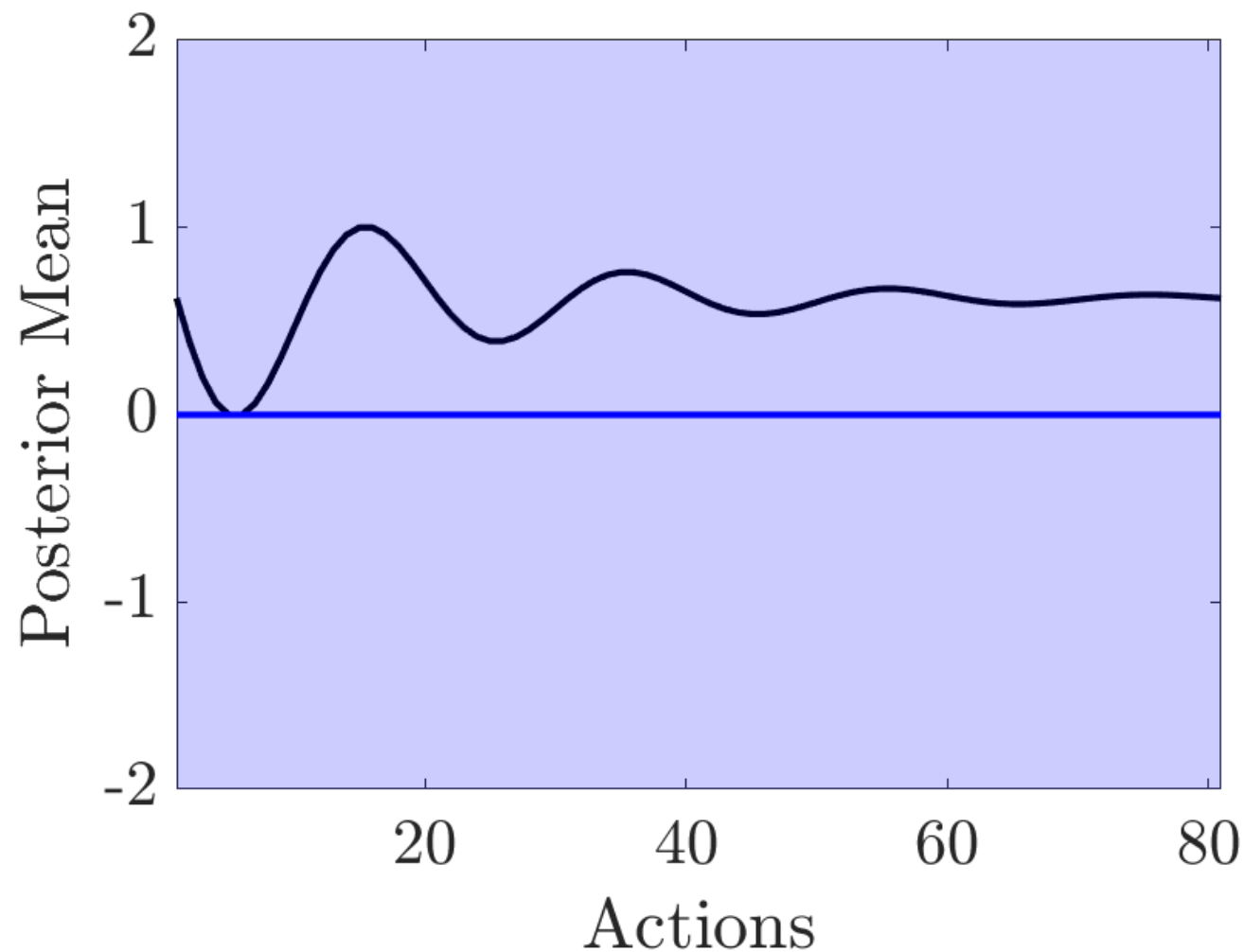
Select actions to: Maximize final utility
(Upper Confidence Bound, Thompson Sampling, Max-Value Entropy Search)

E.g., protein engineering, personalization



How to Select the Next Action

□ Example of Bayesian optimization through Thompson Sampling



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Motivation: Exploration in RL v.s. Human Exploration

- Typically, RL requires very **dense rewards**!



[Mnih *et.al.* , Nature 2015]



[Jaderberg *et.al.*, ICLR 2017; Mirowski *et.al.*, ICLR 2017]

Motivation: Exploration in RL v.s. Human Exploration

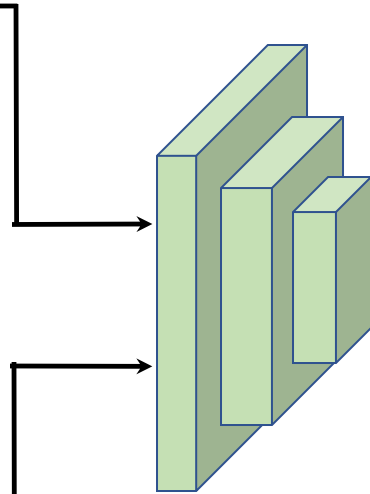
- ❑ Policy gradient for tasks with sparse rewards:



current x_t



goal x_G

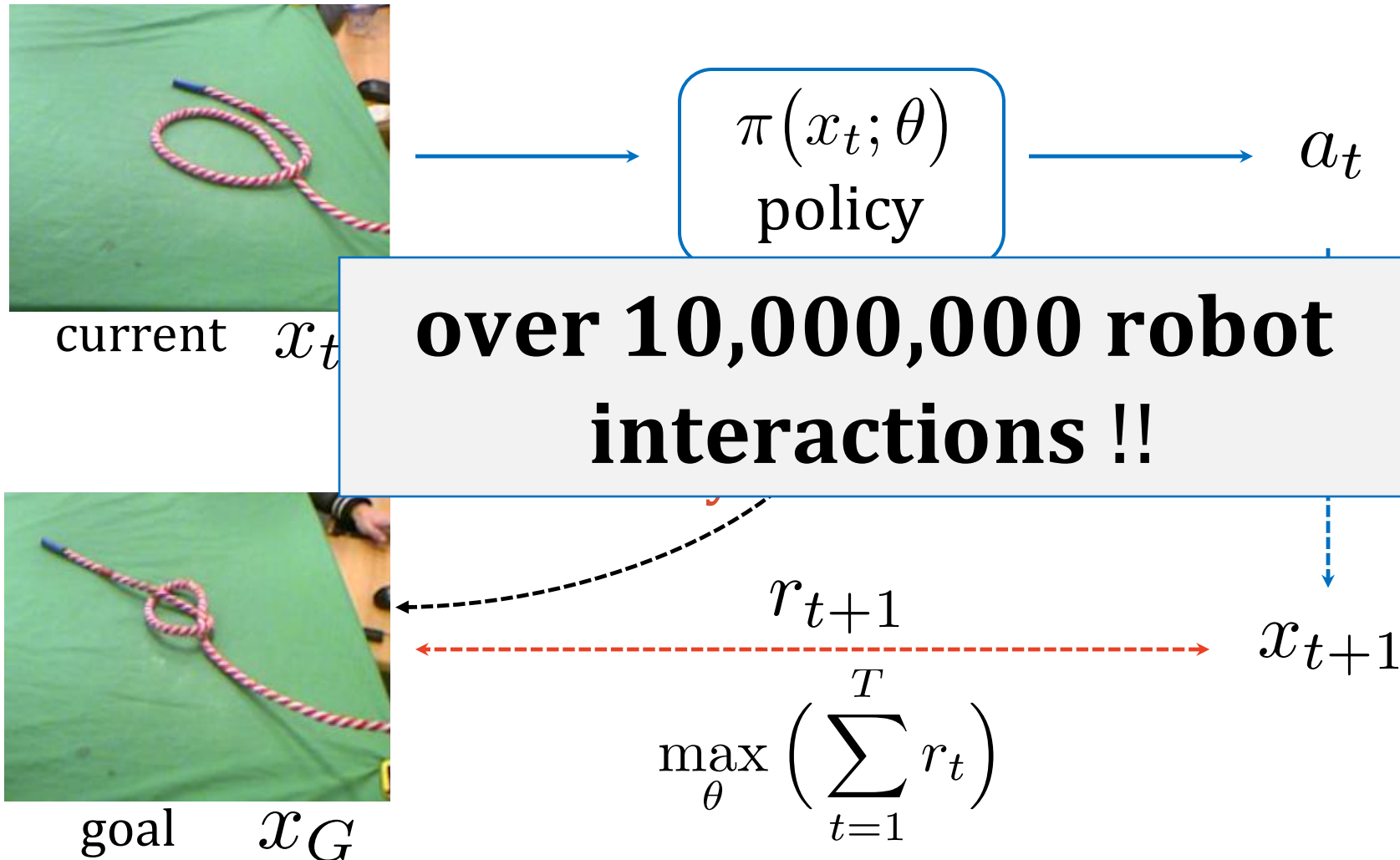


→ actions?



Motivation: Exploration in RL v.s. Human Exploration

- ❑ Policy gradient for tasks with sparse rewards:



Motivation: Exploration in RL v.s. Human Exploration

- ❑ Human babies are very good at explorations!
- ❑ They have “intrinsic motivation” or “curiosity”



[The Scientist in a Crib, Gopnik, Meltzoff & Kuhl, 1999]
[Six Lessons from Babies, Smith & Gasser, 2005]

[Classic definitions and new directions, Ryan *et.al.*, 2000]
[Curiosity and Motivation, Silvia, 2012]

Motivation: Exploration in RL v.s. Human Exploration



Instead of trying to produce a programme to simulate the adult mind, why not rather try to produce one which simulates the child's? If this were then subjected to an appropriate course of education one would obtain the adult brain.

— *Alan Turing* —

AZ QUOTES

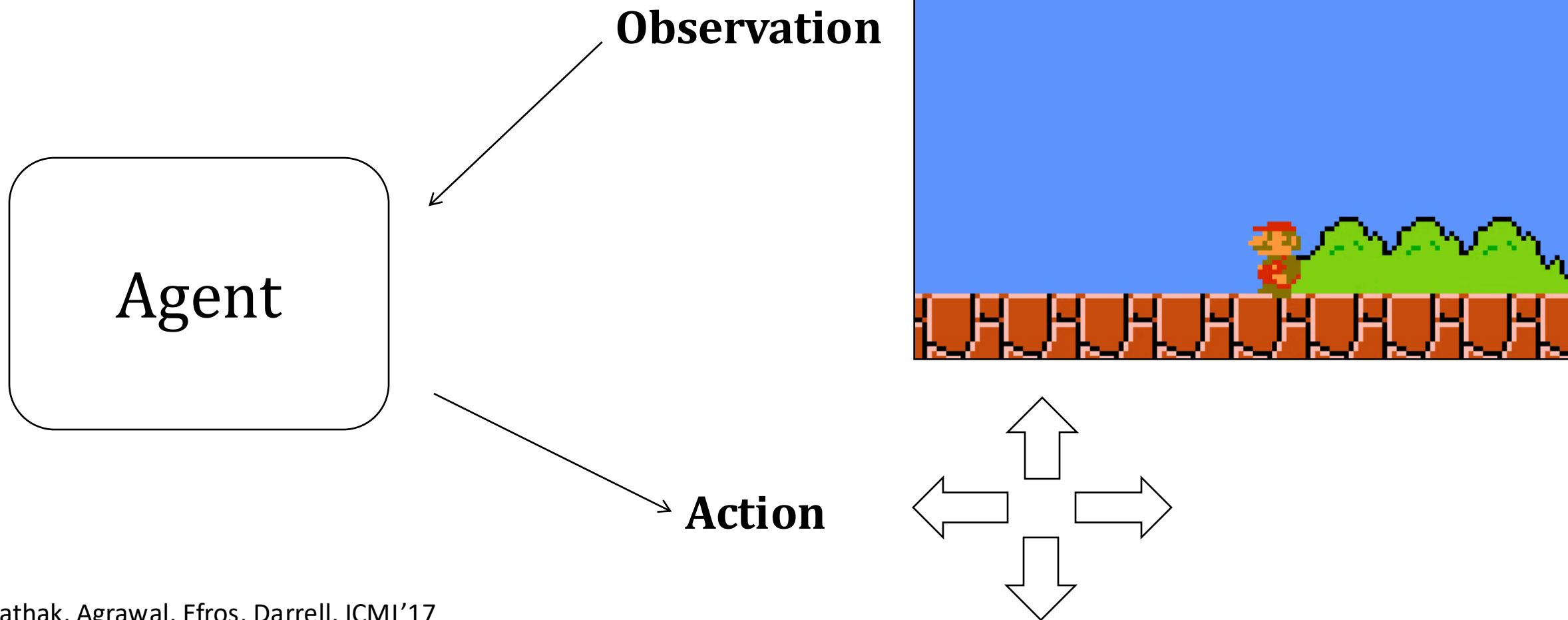
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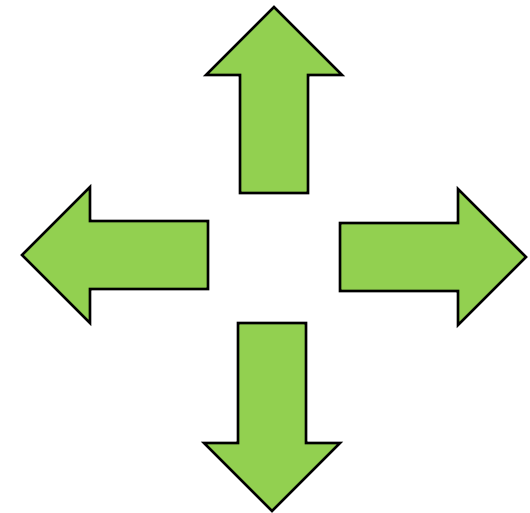
Curiosity

- ❑ Conjecture: exploring **without any external goal** allows the agent to generalize better
- ❑ Example: Super Mario Bros



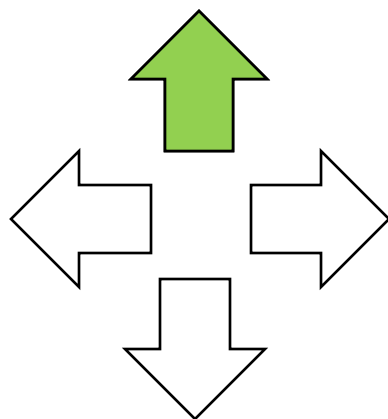
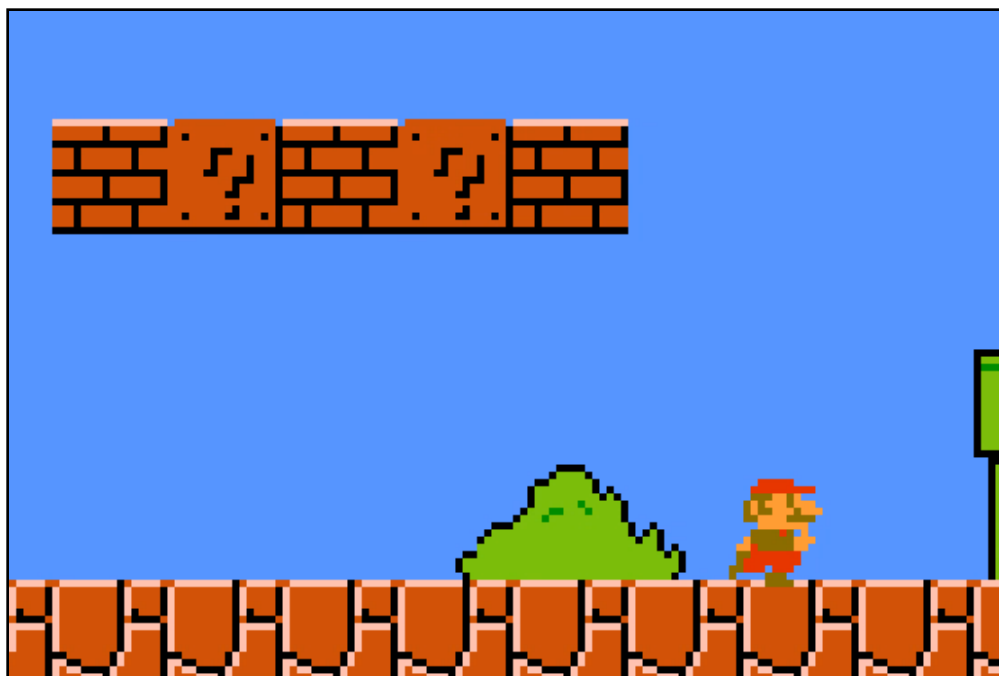
Curiosity

- ❑ Algorithm has learned: all actions have no effect

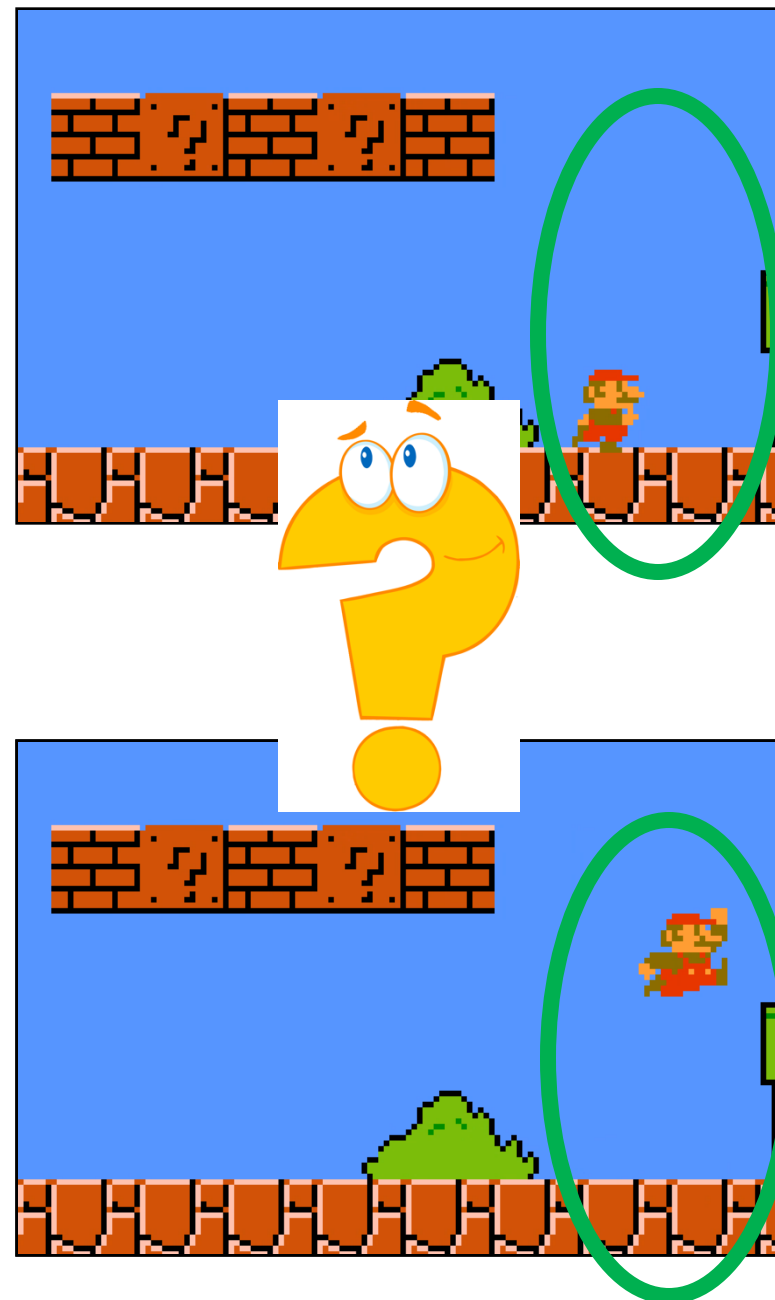


“Down” has no effect

Curiosity

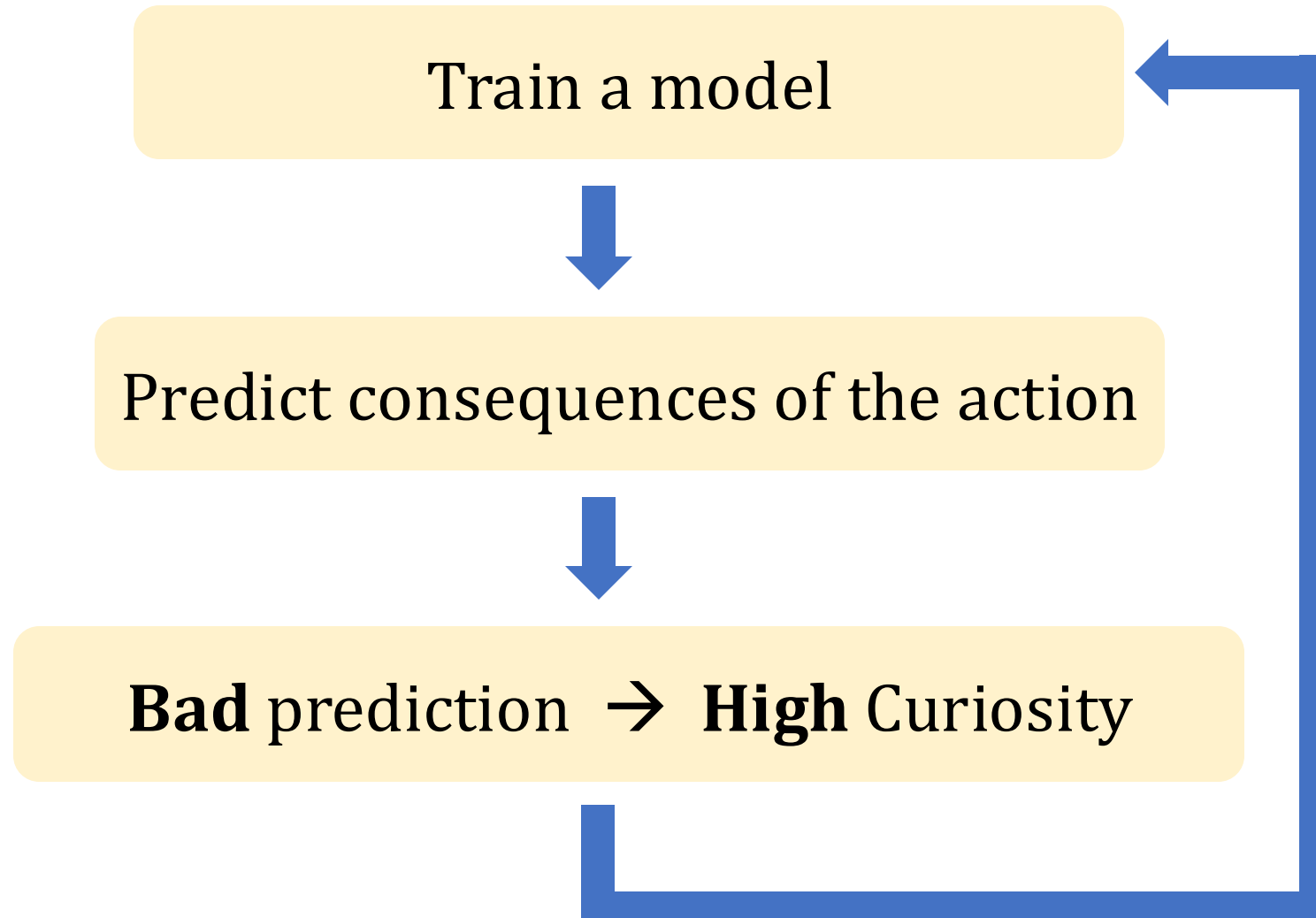


Prediction



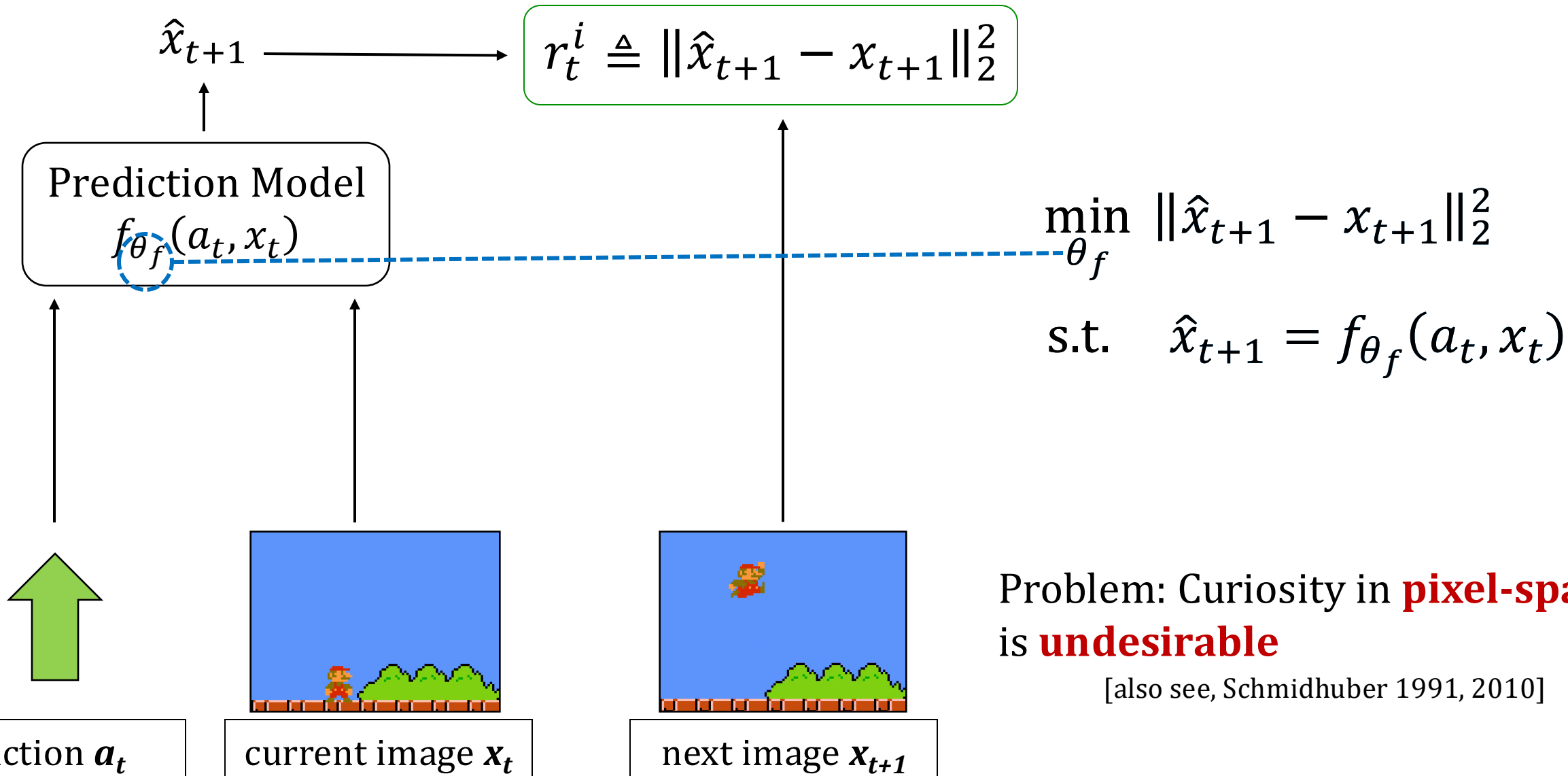
Curiosity

□ Key idea: **Curiosity** $\triangleq$ **Prediction Error**

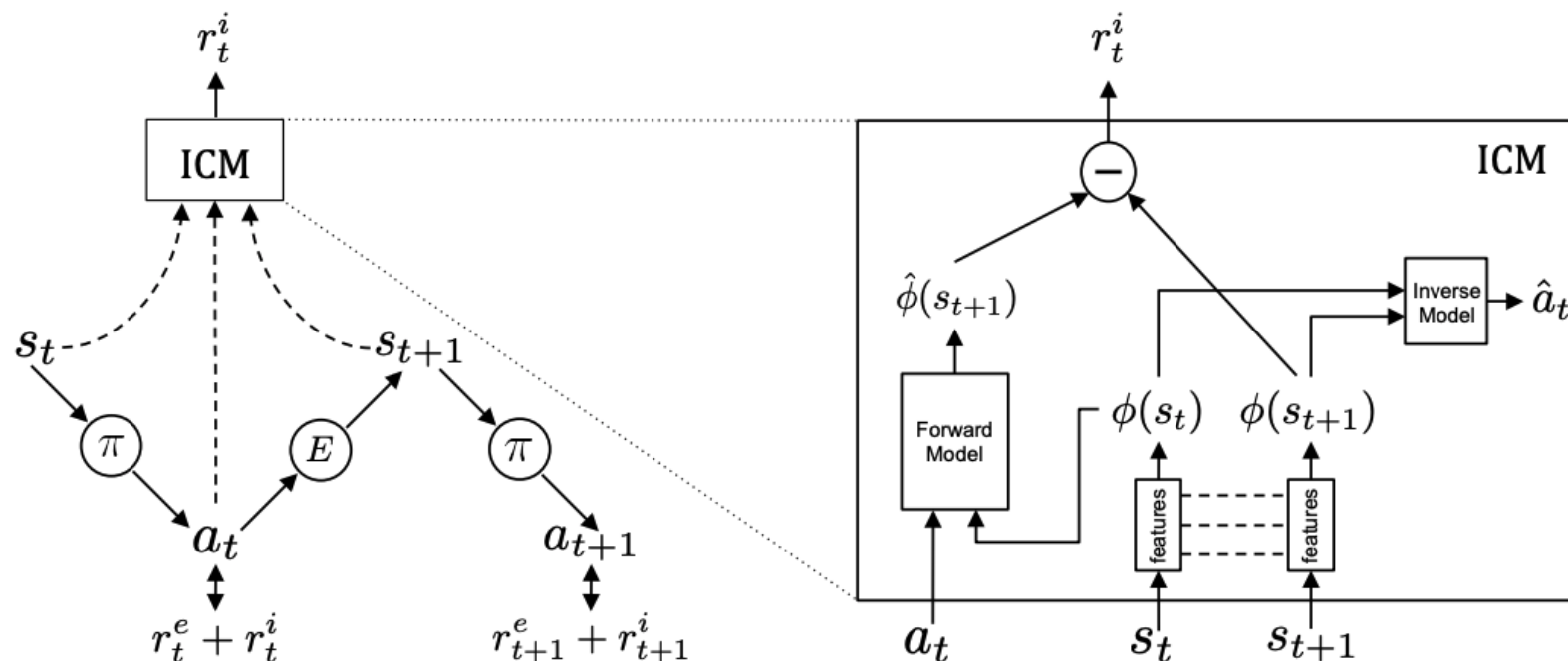


Curiosity

Curiosity Reward

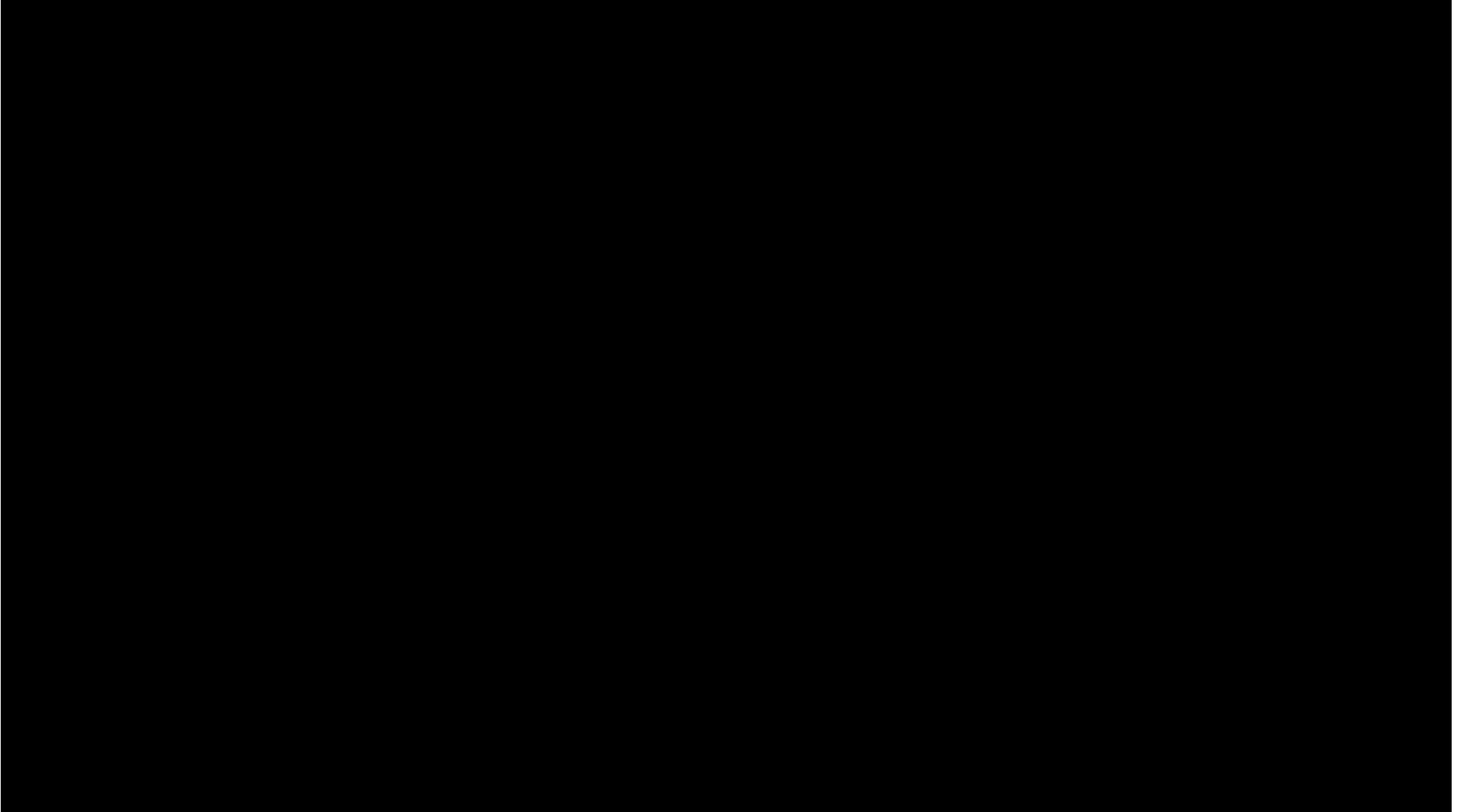


Curiosity



- ❑ r^e : extrinsic, goal-related reward; r^i : intrinsic, curiosity reward
 - $r^e + r^i$ balances between exploration and exploitation (like UCB!)
- ❑ Learning dynamics in feature space: predict $\phi(s_{t+1})$ given $\phi(s_t)$ and a_t
 - Train ϕ using inverse dynamics: the goal is to predict a_t using $\phi(s_{t+1})$ and $\phi(s_t)$

Curiosity



Three Types of Intrinsic Reward

- ❑ Knowledge-based: Surprise / unpredictability / how much learned about world from experience
 - E.g., curiosity

In the following slides:

- ❑ Competence-based: Empowerment / Skills
- ❑ Data-based: Entropy (i.e. coverage) of data collected

Competence-based: Empowerment / Skills

- ❑ $r^i = \log q_\phi(z | \tau) - \log p(z)$
- ❑ Intuition: need to be able to recover the “intent”/“latent” z from the trajectory τ ; i.e., different z results in **distinctly different trajectories**
- ❑ Pipeline:
 - Sample z
 - Train policy conditioned on z , roll out the policy to get trajectories
 - From trajectory, predict z back
 - Reward the policy proportional to how well can you predict z . Also encourage entropy of z to prevent collapse.

Competence-based: Empowerment / Skills

❑ Example: Diversity is all you need (DIAYN)

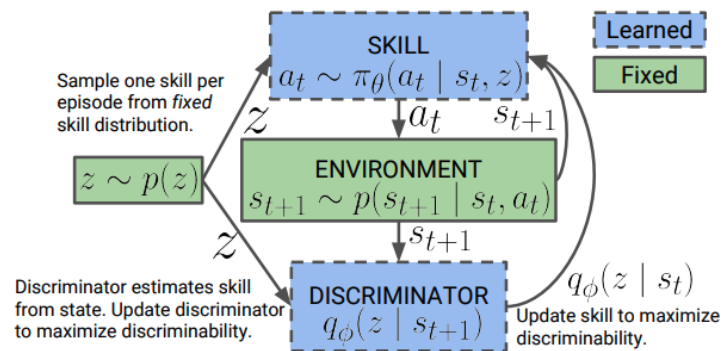
DIVERSITY IS ALL YOU NEED: LEARNING SKILLS WITHOUT A REWARD FUNCTION

Benjamin Eysenbach*
Carnegie Mellon University
beysenba@cs.cmu.edu

Abhishek Gupta
UC Berkeley

Julian Ibarz
Google Brain

Sergey Levine
UC Berkeley
Google Brain



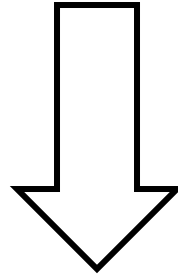
Algorithm 1: DIAYN

```
while not converged do
  Sample skill  $z \sim p(z)$  and initial state  $s_0 \sim p_0(s)$ 
  for  $t \leftarrow 1$  to  $steps\_per\_episode$  do
    Sample action  $a_t \sim \pi_\theta(a_t | s_t, z)$  from skill.
    Step environment:  $s_{t+1} \sim p(s_{t+1} | s_t, a_t)$ .
    Compute  $q_\phi(z | s_{t+1})$  with discriminator.
    Set skill reward  $r_t = \log q_\phi(z | s_{t+1}) - \log p(z)$ 
    Update policy ( $\theta$ ) to maximize  $r_t$  with SAC.
    Update discriminator ( $\phi$ ) with SGD.
```

Data-based: Coverage of Data Collected

- ❑ Key idea: Incentivizing exploration by introducing intrinsic rewards based on a measure of state novelty
- ❑ State entropy as intrinsic reward

$$r^{\text{intrinsic}} = \mathcal{H}(s) = -\mathbb{E}_{s \sim p(s)} [\log p(s)]$$



- ❑ Problem:

Measuring state entropy is intractable to compute in most setting

- ❑ How to fix it: approximate this goal
 - Example (next slide): Random Network Distillation (RND)

Random Network Distillation (RND)

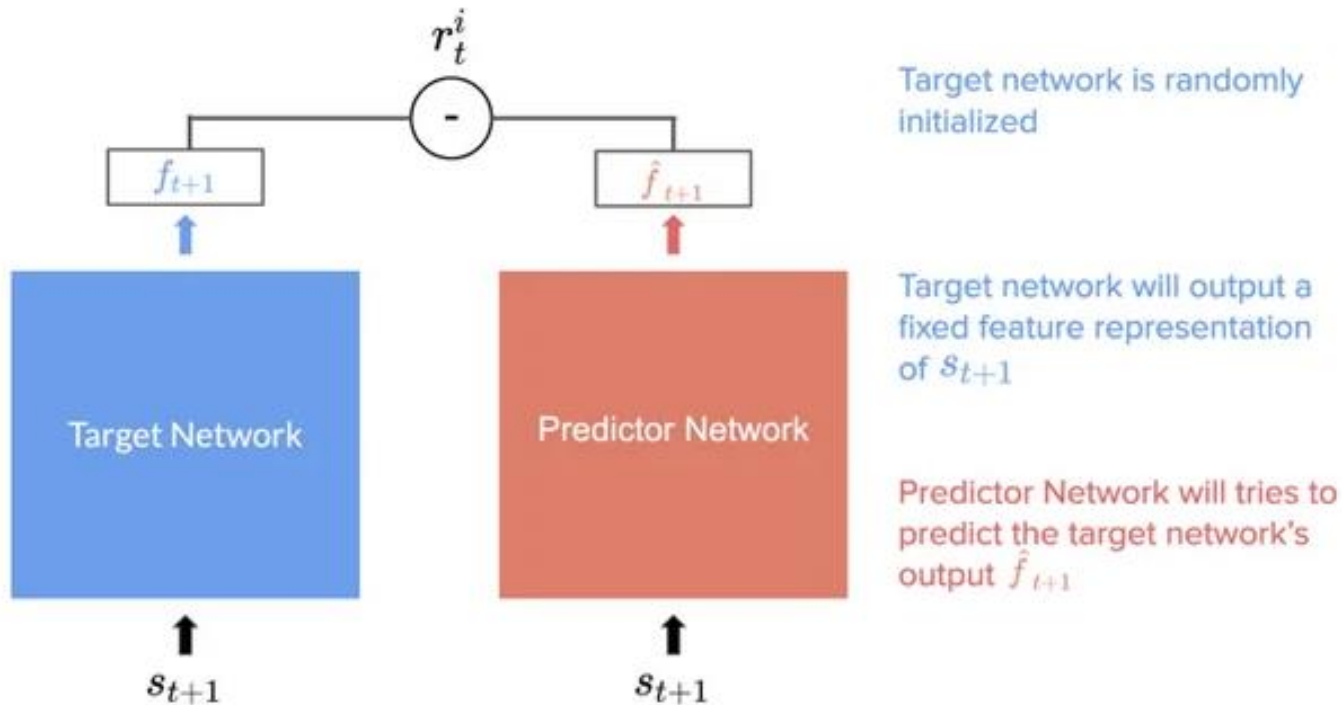
EXPLORATION BY RANDOM NETWORK DISTILLATION

Yuri Burda*
OpenAI

Harrison Edwards*
OpenAI

Amos Storkey
Univ. of Edinburgh

Oleg Klimov
OpenAI

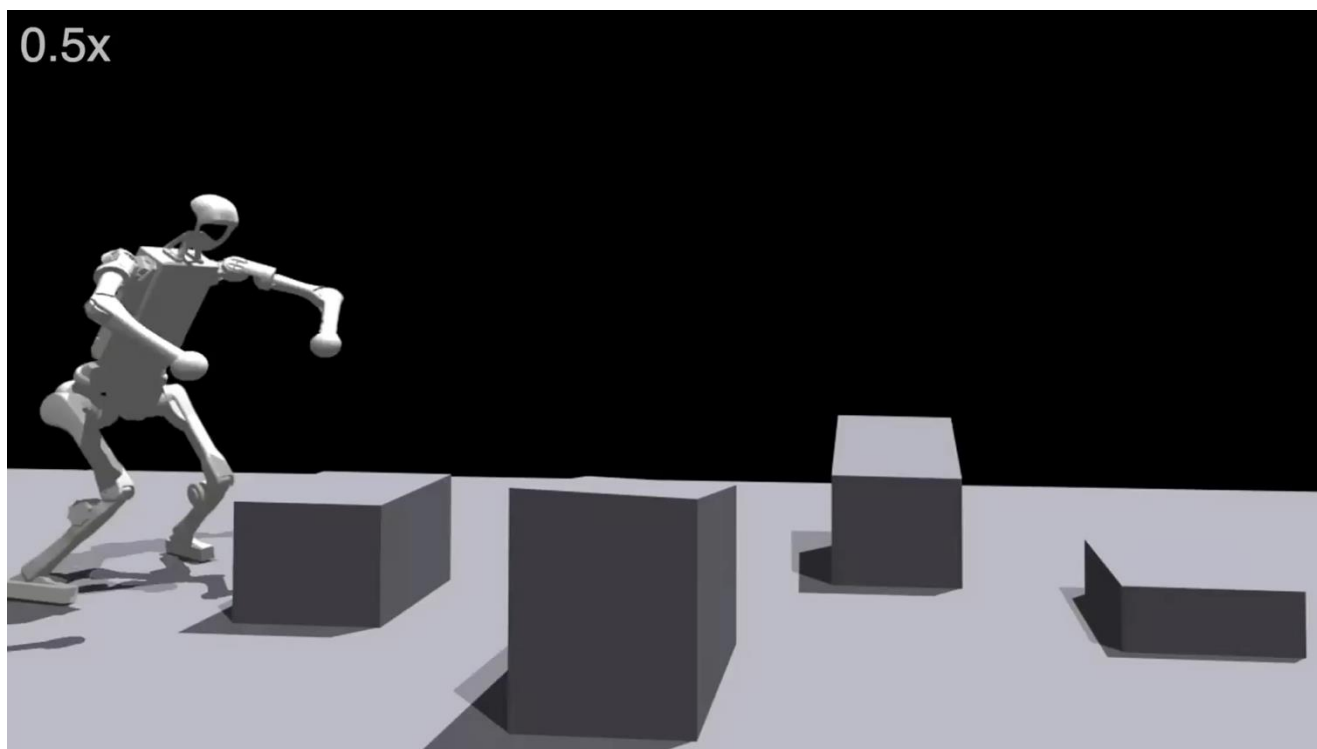


$$r_t^i = \left\| \underbrace{\hat{f}(s_{t+1})}_{\text{Prediction of the Feature Representation of next state}} - \underbrace{f(s_{t+1})}_{\text{Target Feature Representation of next state}} \right\|_2^2$$

L2 norm (to output a scalar)

RND-Style Reward with Hash Encoding

- Key idea: Lower the granularity



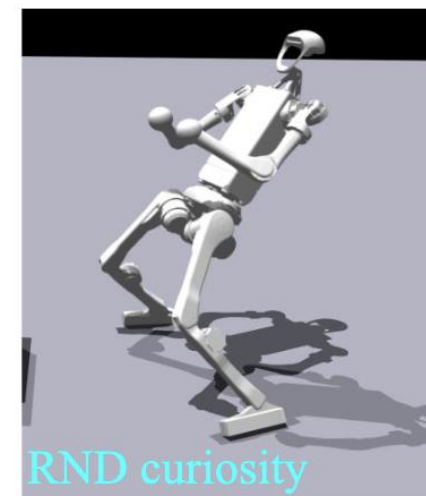
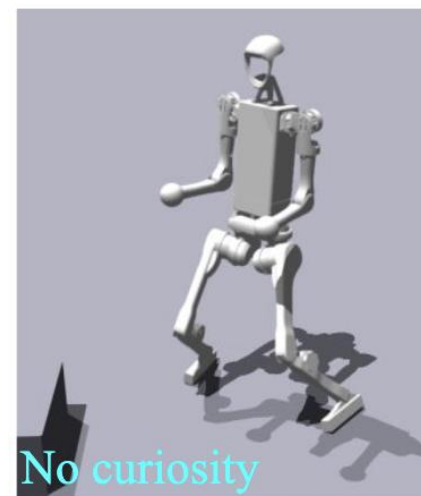
WoCoCo: Learning Whole-Body Humanoid Control with Sequential Contacts

Chong Zhang* Wenli Xiao* Tairan He Guanya Shi

*equal contributions

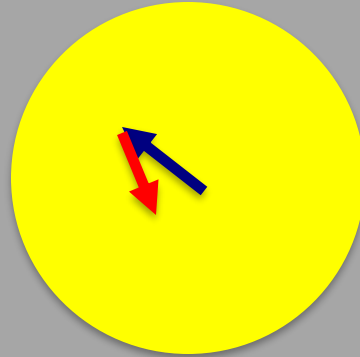
CoRL 2024

$$f_{\text{curi}} : \mathbb{R}^{\dim(o_{\text{curi}})} \rightarrow \mathbb{R}^{\dim(\text{hash})}$$
$$r_{\text{curi}} = \frac{1}{\sqrt{\# \cdot \text{visits of bucket}(o_{\text{curi}})}}.$$



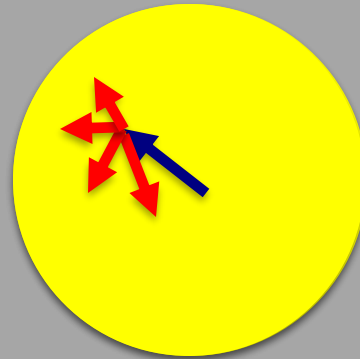
Random Exploration v.s. Novelty Seeking Exploration

Random Exploration



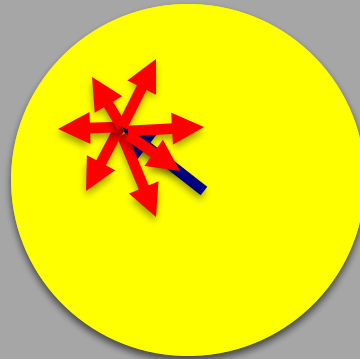
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Random Exploration



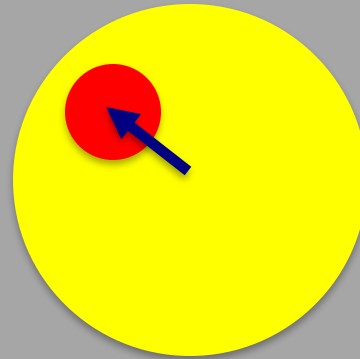
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Random Exploration



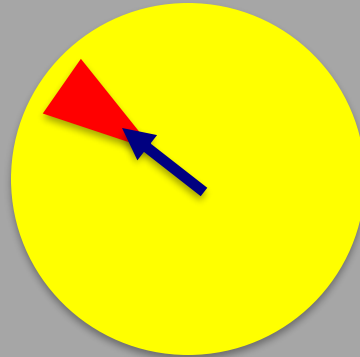
Random Exploration v.s. Novelty Seeking Exploration

Random Exploration



Random Exploration v.s. Novelty Seeking Exploration

Novelty Seeking Exploration



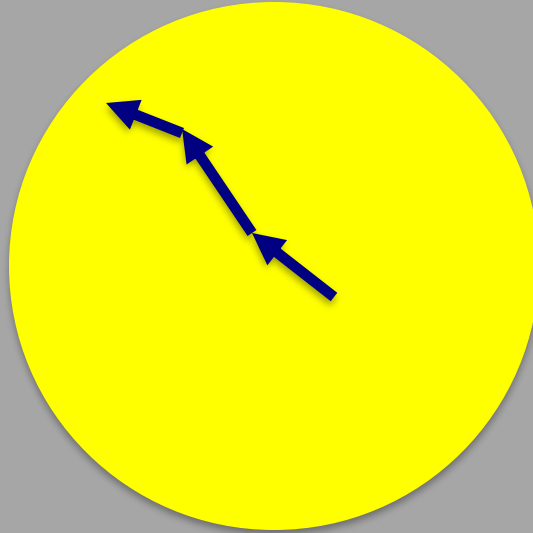
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Novelty Seeking Exploration



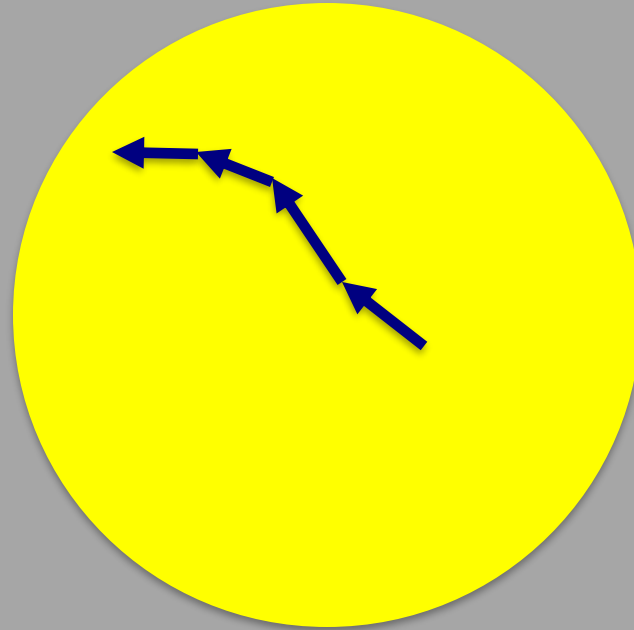
Random Exploration v.s. Novelty Seeking Exploration

Novelty Seeking Exploration



Random Exploration v.s. Novelty Seeking Exploration

Novelty Seeking Exploration



Thank You!